

CRETE-CHEM — Approved UI/UX Theme Migration Specification

Purpose: Migrate the existing functional Next.js + MongoDB application so its public website and admin experience visually and structurally align with the approved HTML reference, without breaking the existing backend, database, cart, enquiry, recommendation, or admin functionality.

Implementation rule: Approved HTML is the visual/source-of-truth reference. The current Next.js codebase is the functional/technical source of truth.

1. Source Files / Required Inputs

AI builder must receive the current codebase plus the approved HTML reference before implementation. The HTML reference is required because it contains the exact visual language, section composition, spacing, typography, card treatments, navigation pattern, responsive behavior, and approved layout direction.

Attach these with the PDF:

1. Approved .html file(s) — mandatory visual reference.
2. Current Next.js codebase/repository — mandatory implementation source.
3. Any approved image/asset folder used by the HTML reference, if separate.
4. Client-approved screenshots only where an exact visual state is important.

If the AI builder cannot inspect the HTML file: also attach a ZIP containing the HTML, CSS, JS and assets used by that reference. Do not rely on screenshots alone.

2. Existing Architecture — Preserve

The current application already has a working Next.js public site, MongoDB catalog/data layer, API routes, cart/order flow, inspection/enquiry flow, recommendation assistant and admin area. These are not to be replaced with static HTML.

Preserve: MongoDB connection; catalog repository; products/services/banners/cases/inquiries/media/orders; authentication/session logic; API routes; cart persistence; checkout/order submission; inspection form; WhatsApp continuation; recommendation logic; admin CRUD and media handling.

Do not rebuild the backend just to match the theme. Change data models only where the approved design introduces a genuinely new content type that cannot safely map to the existing model.

3. Target Homepage Structure

#	Target section	Implementation direction
01	Topbar + Header	Redesign; preserve links/actions and responsive menu.
02	Hero	Rebuild visual composition; keep dynamic stats/data.
03	Services	DB-driven six-service cards, new approved styling.
04	Service Impact	New section; service selector + technical details + before/after.
05	Products	DB-driven product families; approved card/grid treatment.
06	Why Choose Us	Visual redesign; retain verified trust content only.
07	Process	Redesign existing process into approved technical/timeline treatment.
08	Industries	Approved industry card treatment and clean anchor/navigation.
09	Completed Projects	Portfolio grid; separate from illustrative before/after cases.
10	Main CTA	New full-width CTA with WhatsApp + inspection actions.
11	Certifications	Approved standards/quality card layout; verify claims.
12	Contact	Redesign only; preserve enquiry submission/API.

#	Target section	Implementation direction
13	Footer	Full approved multi-column footer redesign.
14	Assistant + Cart	Keep functionality; fully align visual shell.

4. File-by-File Change Map

File / area	Priority	Required work
src/app/globals.css	HIGH	Make this the primary design system: colors, typography, spacing, surfaces, borders, buttons, responsive tokens.
src/components/site-header.tsx	HIGH	Topbar, nav, CTA, sticky header, mobile menu and active states.
src/components/sections.tsx	HIGH	Hero, trust, process, industries, certifications, footer and shared sections.
src/components/products.tsx	HIGH	Service/product cards and approved grid/card treatment.
src/components/projects.tsx	HIGH	Before/after presentation; keep existing interaction.
src/components/contact.tsx	MEDIUM	Visual redesign only; preserve API/enquiry behavior.
src/components/assistant.tsx	MEDIUM	Redesign launcher/modal/drawer; preserve recommendation logic.
src/components/admin.tsx	HIGH	Approved admin visual shell while preserving CRUD/API behavior.
src/components/store-provider.tsx	MEDIUM	Compatibility check for cart/image/state; avoid unnecessary logic changes.
src/components/ui.tsx	MEDIUM	Shared modal, buttons, headings and interaction primitives.
src/app/(site)/page.tsx	HIGH	Reorder/combine sections to final approved homepage structure.
src/app/(site)/products/*	MEDIUM	Apply the same design system to catalog/detail pages.
src/app/layout.tsx	MEDIUM	Font, metadata and theme-color alignment.
src/lib/seed.ts	REVIEW	Change only after explicit content/data approval.
src/lib/db.ts / repository / auth / API	PROTECT	Do not alter for visual migration unless a verified requirement demands it.

5. Design System Requirements

Use one coherent production design system. The old static styles.css/theme.css must not compete with the React design system. First identify whether legacy CSS is referenced at runtime; then remove, isolate or neutralize duplicate rules safely.

Target visual language: deep navy canvas; cyan primary accent; blue secondary accent; restrained orange/amber accent; blue-gray muted text; thin technical borders; controlled corner radius; premium industrial/engineering feel; strong whitespace; responsive card grids; subtle hover/motion rather than excessive effects.

Typography: align production typography with the approved HTML. Do not keep conflicting Arial/legacy heading rules if the approved reference specifies another font. Load fonts in a maintainable Next.js way rather than relying on duplicated CSS imports.

6. Section-Level Acceptance Criteria

Header: approved topbar information, navigation hierarchy, active state, CTA and mobile behavior.

Hero: approved composition, readable overlay, CTA hierarchy, benefit row and dynamic stats.

Services: six DB-driven services, consistent cards, hover/focus states, inspection action.

Service Impact: interactive service selector + technical content + before/after slider; no duplicate backend logic.

Products: DB-driven content with approved visual treatment; do not hard-code production catalog data unnecessarily.

Projects: distinguish real completed projects from illustrative system studies.

Contact: existing form still submits and returns its normal success/error state.

Assistant: recommendation and WhatsApp flows continue working.

Cart: add/remove/quantity/checkout/order submission continue working.

Admin: existing CRUD, authentication and media functions continue working while the interface matches the approved design

language.

7. Data & Content Mapping Rules

Current product families include Membranes, Coatings, Sealants, Grouts, Admixtures and Repair Solutions. Current services include Roof Waterproofing, Water Tank Leakage, Bathroom Leakage, Basement Seepage, Wall Dampness and Heat Proofing. These are live catalog concepts and must not be overwritten merely because the prototype uses different category wording.

If the approved HTML introduces categories such as Concrete, Waterproofing, Refurbishment, Industrial Flooring, Roofing, or Sealing & Bonding, create a mapping proposal first. Do not silently mutate MongoDB records.

Completed Projects are a different content type from illustrative before/after cases. If genuine project records are required, prefer a dedicated projects model/collection rather than mislabeling existing illustrative cases.

8. Images & Assets

Do not paste large base64 image blobs into production React components. Extract approved visual assets into public/assets or the existing media system, then reference them through the application's image/data architecture. Preserve Next Image optimization where appropriate.

If the approved HTML uses external images, replace them with approved/local production assets where licensing and reliability require it. Do not introduce copyrighted or unverifiable assets simply to reproduce a screenshot.

9. Admin Migration

The static HTML admin is a visual reference only. The current React admin is functional and must remain the operational admin. Restyle its sidebar, top area, stats, tables, panels, editors, modals and responsive behavior to match the approved visual language. Never replace live CRUD with fake prototype numbers or static cards.

10. Legacy / Technical Cleanup — Separate Phase

The project contains legacy/static implementation artifacts and a Drizzle/SQLite schema alongside the runtime MongoDB architecture. Do not mix cleanup of these artifacts into the first visual migration unless a build/runtime issue requires it. After theme migration, perform a separate dependency/reference audit and remove obsolete code only when proven unused.

11. Responsive Requirements

Test at minimum: 360px mobile, 390px mobile, 768px tablet, 1024px tablet/small desktop, 1440px desktop and a wide desktop. Verify no horizontal overflow, clipped text, broken grids, inaccessible controls, oversized hero, unusable drawers, or admin sidebar problems. Preserve reduced-motion accessibility behavior.

12. QA / Regression Gate

Before declaring completion: run TypeScript check, lint, production build, route smoke tests, catalog loading, product detail, service interaction, before/after slider, enquiry submission, assistant recommendations, WhatsApp links, cart persistence, order submission, admin login, admin CRUD, media handling, mobile menu and responsive layouts.

Critical rule: visual approval is not complete if any existing functional flow breaks.

13. AI Builder Instructions

Use this order: **inspect** → **map** → **implement** → **validate** → **report**. First inspect the repository and attached HTML. Create a file/change map. Then implement the design system and public site section-by-section. Reuse existing data and business logic. Do not invent APIs or replace working backend behavior. After implementation, run checks and report every changed file plus any unresolved content/data decisions.

When uncertain whether something is visual or functional, preserve the existing functional behavior and change only the presentation until the requirement is explicitly confirmed.

14. Definition of Done

The migration is complete only when the production Next.js site visually aligns with the approved HTML reference across desktop and mobile, while MongoDB data, API routes, cart, enquiries, assistant, WhatsApp continuation and admin functionality remain operational. No fake production data, duplicated design systems, broken legacy routes or unverified claims may be introduced.

15. Attachment Checklist

Attach	Required?
This PDF specification	YES
Approved .html reference	YES — strongly recommended / mandatory for exact visual matching
HTML reference CSS + JS + assets	YES if separate from HTML
Current source repository / ZIP	YES
Approved screenshots	Optional but useful for exact states
Client-approved images/logos	Attach if not already in repository

Recommended handoff package

```
CRETE-CHEM-Migration/  
01_APPROVED_SPEC.pdf  
02_APPROVED_REFERENCE.html  
03_REFERENCE_ASSETS/  
04_CURRENT_CODEBASE.zip  
05_APPROVED_SCREENSHOTS/ (optional)
```

This package gives Codex, Claude Code, Gemini or another AI builder both the what (PDF specification) and the visual truth (.html reference), while the repository provides the how (existing functional architecture).